OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester II Examinations 2023-2024

Course instance codes	3BA1, 4BMS2, 4BS2, 4BME1, 1EM1
Module	Advanced Group Theory
Module code	MA4344
Paper no.	1
External examiner	Prof. Colva Roney-Dougal
Internal examiners	Dr Niall Madden
	Prof. Dane Flannery*
Instructions	Answer all questions; show all working
Duration	2 hours
No. Pages	3 pages (including this cover page)
School	Mathematical and Statistical Sciences
Discipline	Mathematics

Requirements

Release in exam venue	Yes
Release to Library	Yes
Log tables	No
Other materials	No

1. (a) For $m \in \mathbb{Z}$, define x_m to be the 2×2 matrix $\begin{pmatrix} 1+m & m \\ -m & 1-m \end{pmatrix}$. Let S be the set of all x_m as m ranges over $\mathbb{Z}$. Prove that S is a subgroup of $\mathrm{SL}(2,\mathbb{Z})$ isomorphic to $\mathbb{Z}$. **9 marks**

(b) (i) State (but do not prove) Lagrange's theorem and the fundamental isomorphism theorem for groups. **6 marks**

(ii) Let $\alpha\colon G \to H$ be a homomorphism between finite groups G, H. Prove that $|\alpha(G)|$ divides both $|G|$ and $|H|$. **6 marks**

(iii) Using part (ii), explain why the only homomorphism between a group of order 357 and a group of order 437 is the trivial homomorphism. **4 marks**

2. (a) Let G be a group, and let $H \leq G$.

(i) Prove that the number $|G : H|$ of distinct left cosets of H in G is equal to the number of distinct right cosets of H in G. **5 marks**

(ii) Prove that if $|G : H| = 2$ then H is a normal subgroup of G. **7 marks**

(b) Let G be a group acting on a set X. Give a homomorphism $\phi\colon G \to \mathrm{Sym}(X)$ with kernel $\cap_{x\in X} G_x$, where G_x denotes the stabilizer of x under the action of G. **5 marks**

(c) Let G be a group with a subgroup H such that $|G : H| = 3$. Prove that H contains a normal subgroup N of G such that $|G : N|$ divides 6. **8 marks**

3. (a) Demonstrate an action of $\mathrm{Sym}(3)$ on the set of its elements, and use that action to construct an explicit isomorphism of $\mathrm{Sym}(3)$ onto a subgroup of $\mathrm{Sym}(6)$. **9 marks**

(b) Let G be a nontrivial finite group.

(i) Derive the class equation: $|G| = |Z(G)| + \sum_{i=1}^{t} |G : C_G(y_i)|$ where $y_1, \ldots, y_t$ are representatives of the distinct conjugacy classes of G not in $Z(G)$, and $C_G(y_i) = \{x \in G \mid xy_i = y_i x\}$. **6 marks**

(ii) Suppose that G has prime-power order. Using the class equation, prove that $Z(G)$ is nontrivial. **5 marks**

(iii) Using part (ii) and the fact that if $G/Z(G)$ is cyclic then G is abelian, explain why a group of order p^2, p prime, is abelian. **5 marks**

4. (a) Let G be a finite abelian group of order divisible by a prime p. Using induction on $|G|$, prove that G contains an element of order p. **8 marks**

(b) State without proof the three parts (existence, uniqueness, arithmetic) of Sylow's Theorem. **6 marks**

(c) Prove that there is no simple group of order

(i) 12 **6 marks**

(ii) 56. **5 marks**